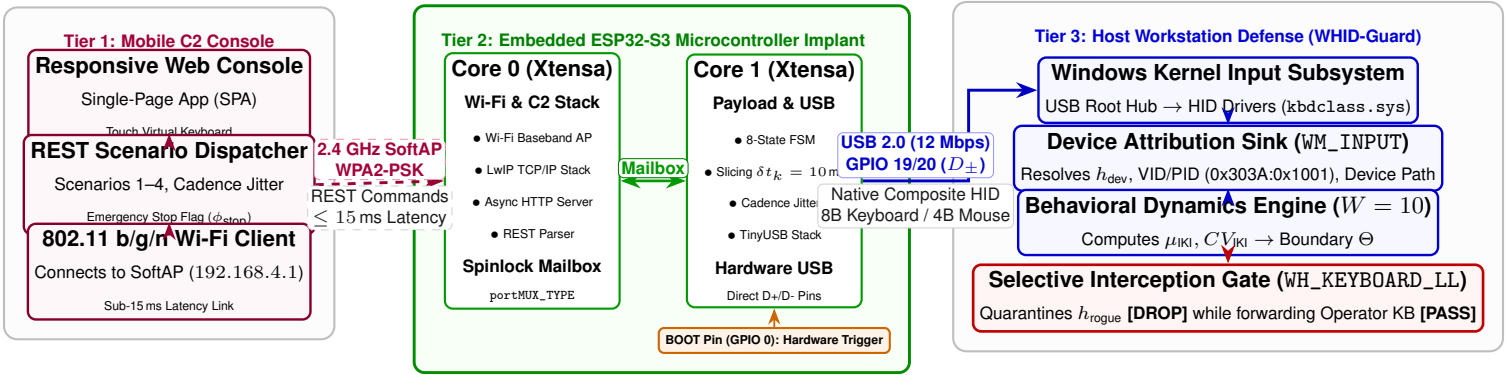

Tier 1: Mobile C2 Console
Responsive Web Console
Single-Page App (SPA)
Touch Virtual Keyboard
REST Scenario Dispatcher
Scenarios 1–4, Cadence Jitter
Emergency Stop Flag (ϕ_{stop})
802.11 b/g/n Wi-Fi Client
Connects to SoftAP (192.168.4.1)
Sub-15 ms Latency Link
2.4 GHz SoftAP
WPA2-PSK
REST Commands
$\leq$ 15 ms Latency
Tier 2: Embedded ESP32-S3 Microcontroller Implant
Core 0 (Xtensa)
Wi-Fi & C2 Stack
Wi-Fi Baseband AP
LwIP TCP/IP Stack
Async HTTP Server
REST Parser
Spinlock Mailbox
portMUX_TYPE
Mailbox
Core 1 (Xtensa)
Payload & USB
8-State FSM
Slicing $\delta t_k = 10$ m
Cadence Jitte
TinyUSB Stack
Hardware USB
Direct D+/D- Pins
BOOT Pin (GPIO 0): Hardware Trigger
USB 2.0 (12 Mbps)
GPIO 19/20 ($D_\pm$)
Native Composite HID
8B Keyboard / 4B Mouse
Tier 3: Host Workstation Defense (WHID-Guard)
Windows Kernel Input Subsystem
USB Root Hub → HID Drivers (kbdclass.sys)
Device Attribution Sink (WM_INPUT)
Resolves h_{dev}, VID/PID (0x303A:0x1001), Device Path
Behavioral Dynamics Engine ($W = 10$)
Computes μ_{IKI}, $CV_{\text{IKI}} \rightarrow$ Boundary Θ
Selective Interception Gate (WH_KEYBOARD_LL)
Quarantines h_{rogue} [DROP] while forwarding Operator KB [PASS]